Status	Finished
Started	Thursday, 29 October 2026, 9:49 PM
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Duration	2 mins 52 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Checking each element sequentially ✓
c. Dividing the list in half repeatedly
d. Skipping every second element

Question **2**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search backtracks to find duplicate elements
c. The search stops immediately ✓
d. The search continues until the end of the list

Question **3**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search works on unsorted lists
d. Binary search can find multiple instances of the target element

Question **4**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Linear search ✓

c. Hash search

d. Binary search

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.

b. Linear search can be applied to both sorted and unsorted lists. ✓

c. Linear search is more efficient than binary search on large lists.

d. Linear search requires the list to be sorted.

Question **6**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search

b. Binary search ✓

c. Sequential search

d. Hash search

Question **7**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted

b. The search continues in the right sublist

c. The search ends successfully ✓

d. The search continues in the left sublist

Question **8**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Discovering
c. Mining
d. Searching ✓

Question **9**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 1000
c. 10 ✓
d. 500

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues in the left or right sublist ✓
b. The search stops
c. The list is sorted
d. The search continues from the beginning

Question **11**

Incorrect

Mark 0.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and sorted
b. When the list is large and sorted
c. When the list is small and unsorted
d. When the list is large and unsorted ✗

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There should be direct access to the middle element in any sublist

c. The list must be sorted

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O👎

b. O(n log n)

c. O(log n) ✓

d. O(1)

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves

b. By using a hash function

c. By sorting the list first

d. By comparing each element in the list sequentially ✓

Question **15**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear

b. Binary ✓

c. Both (1) & (3)

d. Hash